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Are X-ray Landmark Detection Models Fair?
A preliminary assessment and mitigation strategy

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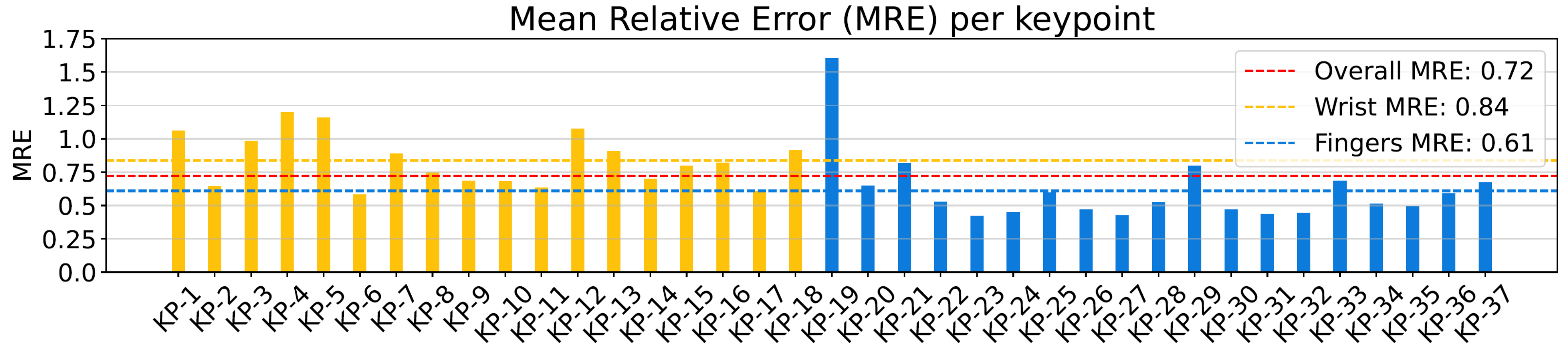
Fairness Assessment

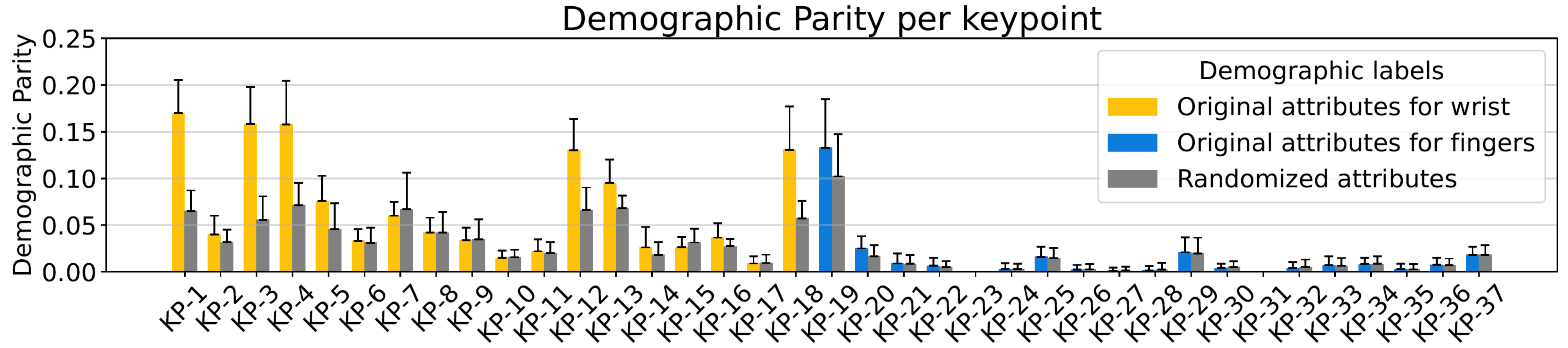
Landmark detection is usually judged by overall accuracy, but this can hide important differences.

We introduce an assessment that looks at each keypoint separately and across demographic groups, showing that fairness must be considered beyond global averages

Mean Relative Error (MRE) per keypoint



Demographic Parity per keypoint



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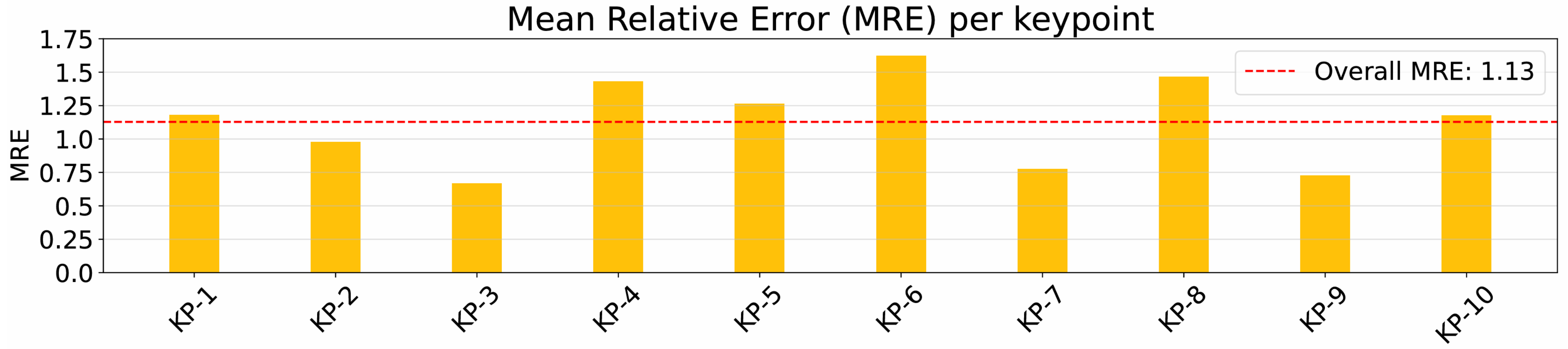
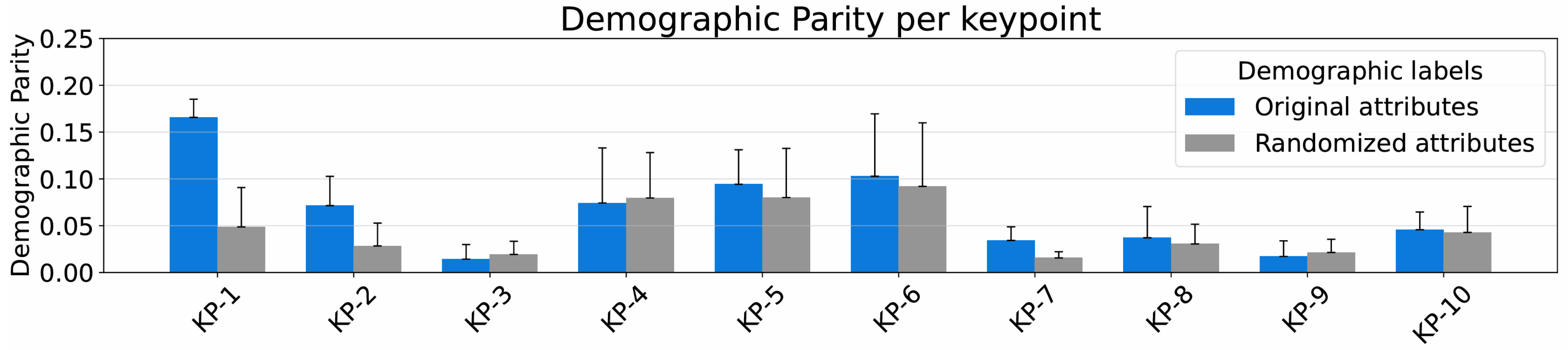


Demographic Parity per keypoint



Adapting Fairness Metrics

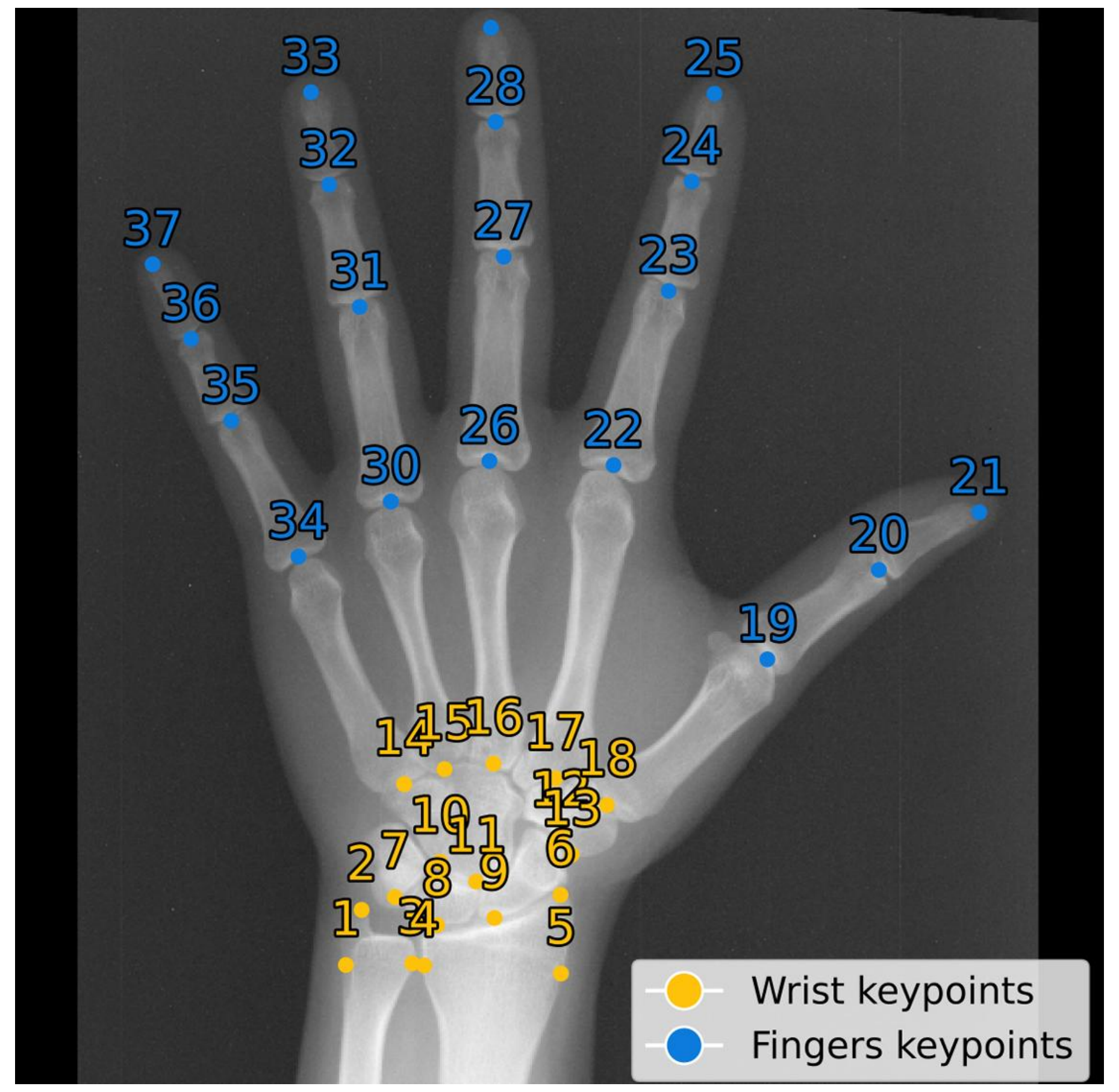
To measure fairness, we adapt Demographic Parity from classification to landmark detection.

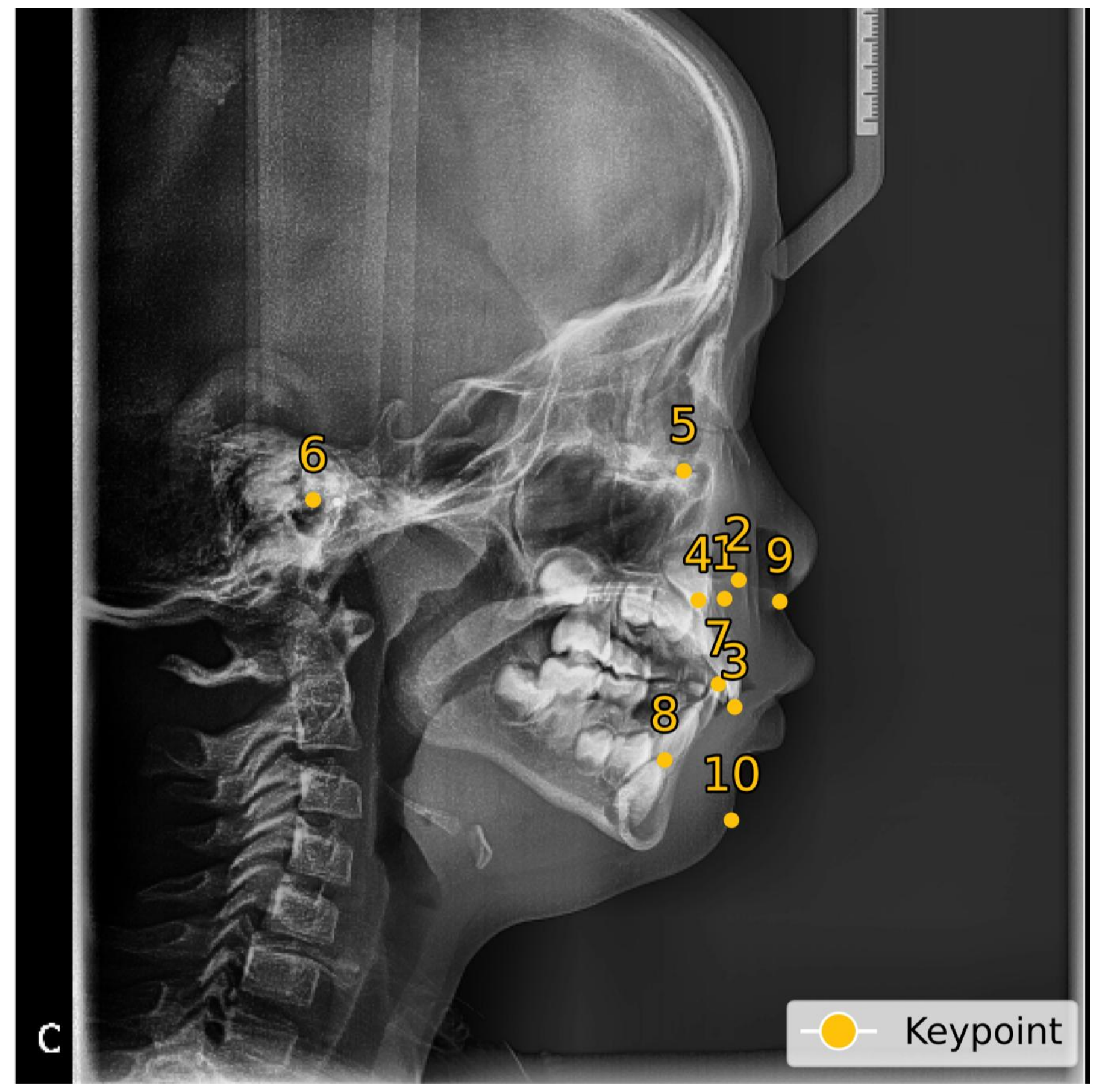
By linking success rates at each keypoint with demographic attributes, we reveal differences that would remain invisible when only looking at aggregated results

$$\underbrace{\Delta_{\mathrm{DP}} = \max_{g_i, g_j \in \mathcal{G}} \left| \Pr\left(\hat{Y} = 1 \mid A = g_i\right) - \Pr\left(\hat{Y} = 1 \mid A = g_j\right) \right|}_{\text{Original Demographic Parity}}$$

$$\Pr\left(\hat{Y} = 1 \mid A = g\right) \longmapsto \mathrm{SDR}(g) = \frac{\#\{\, i \in g : \mathrm{error}(i) \le \tau \,\}}{\#\{\, i \in g \,\}}$$

$$\underbrace{\Delta_{\mathrm{SDR}} = \max_{g_i, g_j \in \mathcal{G}} \left| \mathrm{SDR}(g_i) - \mathrm{SDR}(g_j) \right|}_{\text{Adapted Demographic Parity for Landmark Detection}}$$



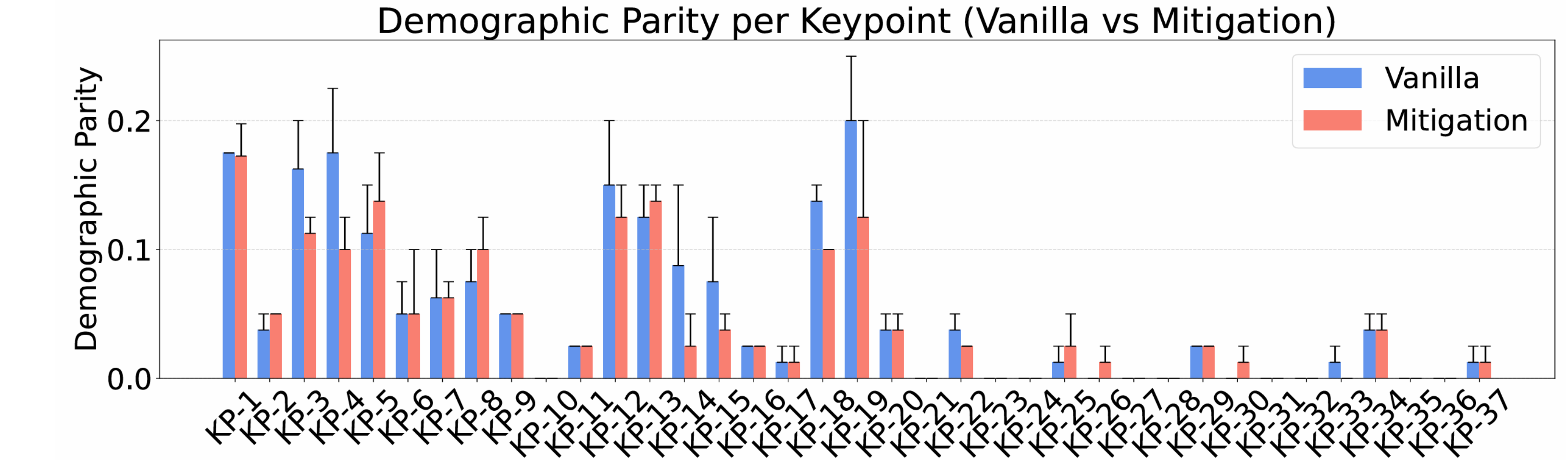


Fairness Mitigation

We explore GroupDRO, a method designed to improve outcomes for underperforming groups in classification tasks.

Applied to landmark detection, it reduces disparities across many keypoints while keeping overall accuracy stable, suggesting a path toward fairer medical AI

Demographic Parity per Keypoint (Vanilla vs Mitigation)



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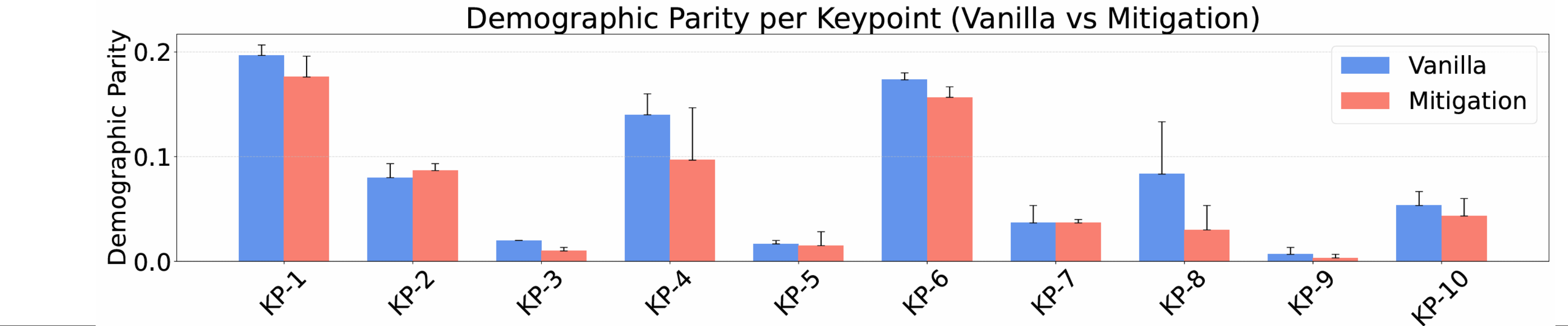

Privacy-related Issues

We also find that errors in landmark prediction can unintentionally expose sensitive patient attributes such as age or gender.

This means that even anonymized datasets may carry privacy risks when fairness is not carefully addressed

Dataset	Sensitive Attribute	Filtered Attribute	CNN image Classifier	RF MRE-based Classifier
DHA	Age	male	0.53 ± 0.08	0.68 ± 0.05
		female	0.56 ± 0.07	0.64 ± 0.12
	Gender	young	0.56 ± 0.07	0.73 ± 0.13
		old	0.55 ± 0.08	0.72 ± 0.07
CephAdoAdu	Age	None	0.59 ± 0.16	0.64 ± 0.05